For **Task 2**,

I need to create a **Smart Agriculture Dashboard Proposal**. Think of it as designing a dashboard that a farmer would use to monitor their farm, crops, weather, and irrigation in one place.

Step-by-Step Guide

Step 1: Create a Title Page

Include:

- KhetHub Internship Program 2026
- Task 2: Smart Agriculture Dashboard Challenge
- Submitted by: Pranshu Tyagi
- Role (Web Development Intern / Backend Developer)
- Submission Date: 24 june 2026

Step 2: Write an Introduction.

Explain what the dashboard does.

Example:

The KhetHub Smart Agriculture Dashboard is a centralized platform that helps farmers monitor crop health, weather conditions, soil moisture levels, irrigation status, and farming activities. The dashboard provides real-time insights and recommendations to improve agricultural productivity.

Step 3: Dashboard Layout Structure

Draw a simple dashboard layout (can be made in Word, PowerPoint, Canva, or Figma).

Sections to Include:

1. Farm Overview

Display:

- Total Farms
- Active Crops
- Total Area
- Irrigation Status

2. Weather Information Panel

Display:

- Temperature
- Humidity

- Rain Forecast
- Wind Speed

3. Crop Health Status

Display:

- Healthy Crops
- Warning Status
- Disease Detection Alerts

4. Soil Moisture Monitoring

Display:

- Current Moisture Level
- Moisture Trend Graph

5. Irrigation Status

Display:

- Pump Status
- Water Usage
- Irrigation Schedule

6. Alerts & Notifications

Display:

- Weather Alerts
- Disease Alerts
- Low Soil Moisture Alerts

7. Task Summary

Display:

- Today's Tasks
- Upcoming Activities
- Completed Tasks

Step 4: Data Visualization

Explain how data will be shown.

Dashboard Cards

Example:

Metric	Value
Temperature	32°C
Humidity	70%
Soil Moisture	55%
Active Crops	4

Charts

Line Chart

- Soil Moisture Trends

Bar Chart

- Weekly Water Usage

Pie Chart

- Crop Health Distribution

Table

- Recent Farm Activities

Step 5: User Experience & Design

Theme

Smart Agriculture Dashboard 🌱

Colors

- Green (#2E7D32)
- Light Green (#81C784)
- White (#FFFFFF)
- Dark Gray (#263238)

Fonts

- Poppins
- Roboto

Design Principles

- Mobile Responsive
- Easy Navigation
- Clear Data Visualization

- Large Readable Text
- Simple Dashboard Cards

Step 6: Technical Approach

For Web Development Intern

Frontend

- HTML
- CSS
- JavaScript
- React.js

Backend

- Node.js
- Express.js

Database

- MySQL

Workflow

```
Farmer Login
      ↓
Frontend Dashboard
      ↓
API Requests
      ↓
Backend Server
      ↓
Database
      ↓
Dashboard Charts & Insights
```

Step 7: Backend Design

Database Schema

Users Table

Field	Type
user_id	INT
name	VARCHAR
email	VARCHAR

Farms Table

Field	Type
farm_id	INT
user_id	INT
farm_name	VARCHAR

Sensor Data Table

Field	Type
sensor_id	INT
moisture	FLOAT
temperature	FLOAT
humidity	FLOAT

Alerts Table

Field	Type
alert_id	INT
message	TEXT
date	DATETIME

Step 8: Sample API

Weather API

```JSON
{
  "temperature": 32,
  "humidity": 70,
  "rainForecast": "20%"
}
```

Soil Data API

```JSON
{
  "soilMoisture": 55,
  "status": "Good"
}
```

Crop Health API

```JSON
{
  "healthy": 85,
  "warning": 10,
  "critical": 5
}
```

Step 9: Bonus Features

Add these to score higher:

- Dark Mode
- AI Crop Recommendations
- Real-Time Sensor Monitoring
- Multilingual Support (Hindi & English)
- Mobile-Friendly Design
- Interactive Charts

Step 10: Conclusion

Example:

The KhetHub Smart Agriculture Dashboard will provide farmers with real-time agricultural insights through weather monitoring, crop health tracking, soil moisture analysis, irrigation management, and intelligent alerts. The platform will improve decision-making, increase productivity, and support modern digital farming practices.

Khethub Smart Agriculture dashboard

🌾 KhetHub Smart Agriculture Dashboard

Temperature	Humidity	Soil Moisture	Active Crops
32°C	70%	55%	4

🌦 Weather Information

Sunny | Rain Chance: 20% | Wind Speed: 12 km/h

🌱 Crop Health Status

Healthy: 85%
Warning: 10%
Critical: 5%

💧 Irrigation Status

Pump Status: ON
Water Usage: 120 Litres

🚨 Alerts

- Low Soil Moisture Alert
- Rain Expected Tomorrow

📋 Today's Tasks

- Water Field A
- Check Crop Health
- Apply Fertilizer

Code 1: Index.HTML

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>KhetHub Smart Agriculture Dashboard</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
    <h1>🌾 KhetHub Smart Agriculture Dashboard</h1>
</header>

<div class="container">

    <div class="cards">
        <div class="card">
            <h3>Temperature</h3>
            <p id="temp">32°C</p>
        </div>

        <div class="card">
            <h3>Humidity</h3>
            <p id="humidity">70%</p>
        </div>

        <div class="card">
            <h3>Soil Moisture</h3>
            <p id="moisture">55%</p>
        </div>

```

Code 2: style.css

```css
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, sans-serif;
}

body{
    background:#f4f7f4;
}

header{
    background:#2e7d32;
    color:white;
    text-align:center;
    padding:20px;
}

.container{
    width:90%;
    margin:20px auto;
```

```
["INFO" - 10:42:46 PM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration
["INFO" - 10:52:31 PM] EditorConfig support is enabled, checking for .editorconfig files
["INFO" - 10:52:31 PM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration
["INFO" - 10:53:12 PM] EditorConfig support is enabled, checking for .editorconfig files
["INFO" - 10:53:12 PM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration
```

Code 3: script.js

```
Welcome    index.html1.html    JS script.js    # style.css 1.css
JS script.js > ...
cons (property) temperature: string
    temperature: "32°C",
    humidity: "70%",
    moisture: "55%"
};

document.getElementById("temp").innerText =
dashboardData.temperature;

document.getElementById("humidity").innerText =
dashboardData.humidity;

document.getElementById("moisture").innerText =
dashboardData.moisture;
```